Prevalence and Characteristics of Asymmetric Hearing Loss in Korean Adults: A KNHANES 2022–2023 Analysis

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Introduction

- **Asymmetric Hearing Loss (AHL)** is defined by interaural threshold difference.
- National epidemiologic data on AHL in Korean adults remain limited.
- This study evaluated prevalence and audiometric characteristics of AHL using **KNHANES 2022–2023**.
- This study aimed to estimate the **prevalence of AHL** in Korean adults aged ≥40 years and to **characterize** its age, frequency, laterality, and interaural difference patterns.

Materials & Methods

- **Definition of AHL**
  Absolute interaural difference ≥15 dB HL in 4-frequency pure-tone average (0.5, 1, 2, 4 kHz), based on the AAO–HNS criterion
- Adults aged **≥40 years**
  - Exclusion: Negative 0.5, 1, 2, and 4 kHz hearing thresholds, type B or flat tympanogram
- Final sample: N = 6574 [Weighted overall population: ~ 22.8 million]
- AHL case subset: n = 447

Results ▶ Key Findings

6.5%
Weighted prevalence of AHL

4 kHz
Most common asymmetry frequency

L > R
Left-sided AHL more common than right

15–20 dB HL
Largest category of interaural difference

Weighted Prevalence of AHL by Age Group (N = 6574)

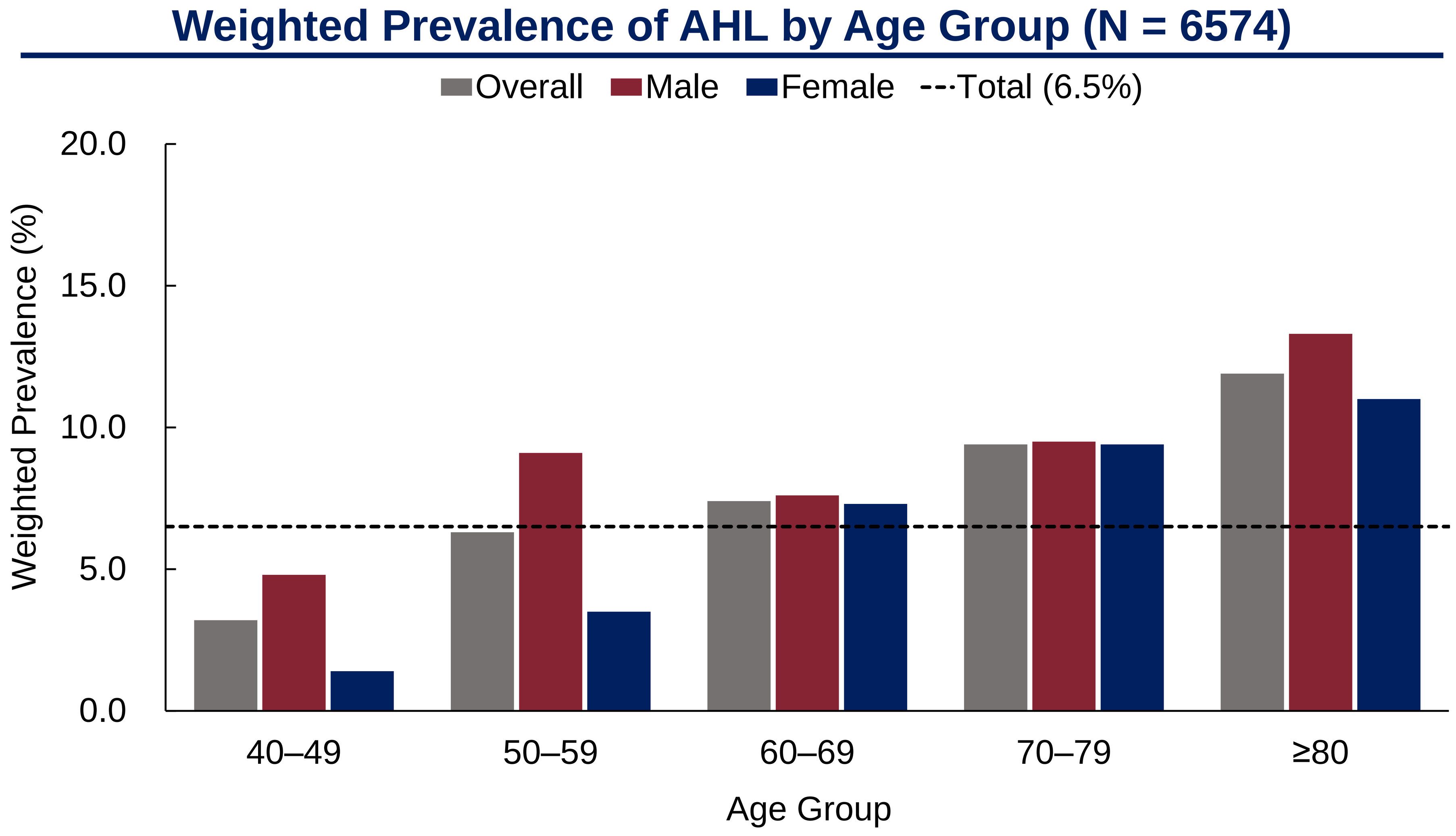

Figure 1. Weighted prevalence of AHL increased with age overall. Men showed a higher prevalence than women. Dashed line indicates the total weighted prevalence of 6.5%

Weighted Mean Hearing Thresholds by Frequency (N = 6574)

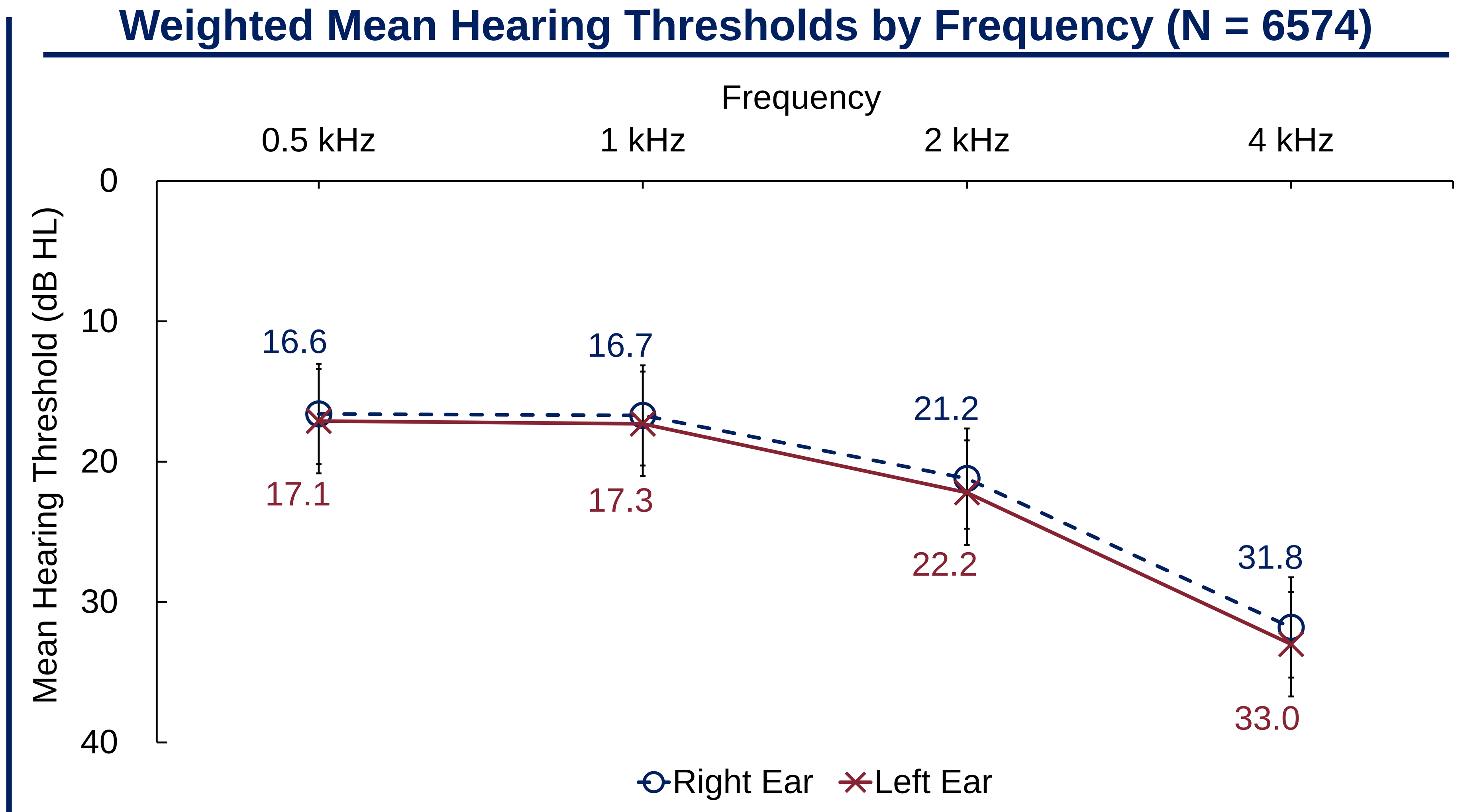

Figure 2. Left-ear hearing thresholds were higher than right-ear hearing thresholds across all tested frequencies, with larger interaural differences at 2 and 4 kHz.

Frequency-Specific Asymmetry (n = 447)

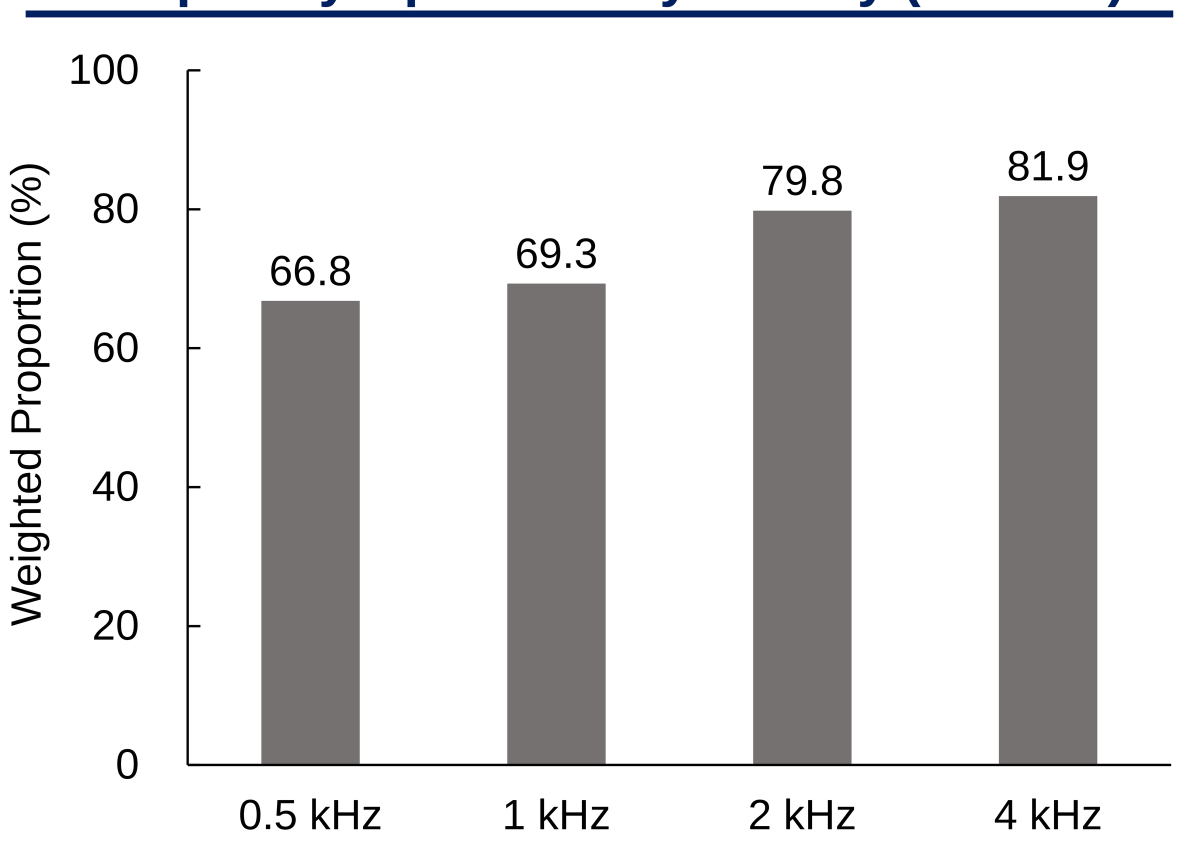

Figure 3. Asymmetry was more common at higher frequencies. Proportions are not mutually exclusive and do not sum to 100%, because participants could have asymmetry at more than one frequency.

Laterality of AHL (n = 447)

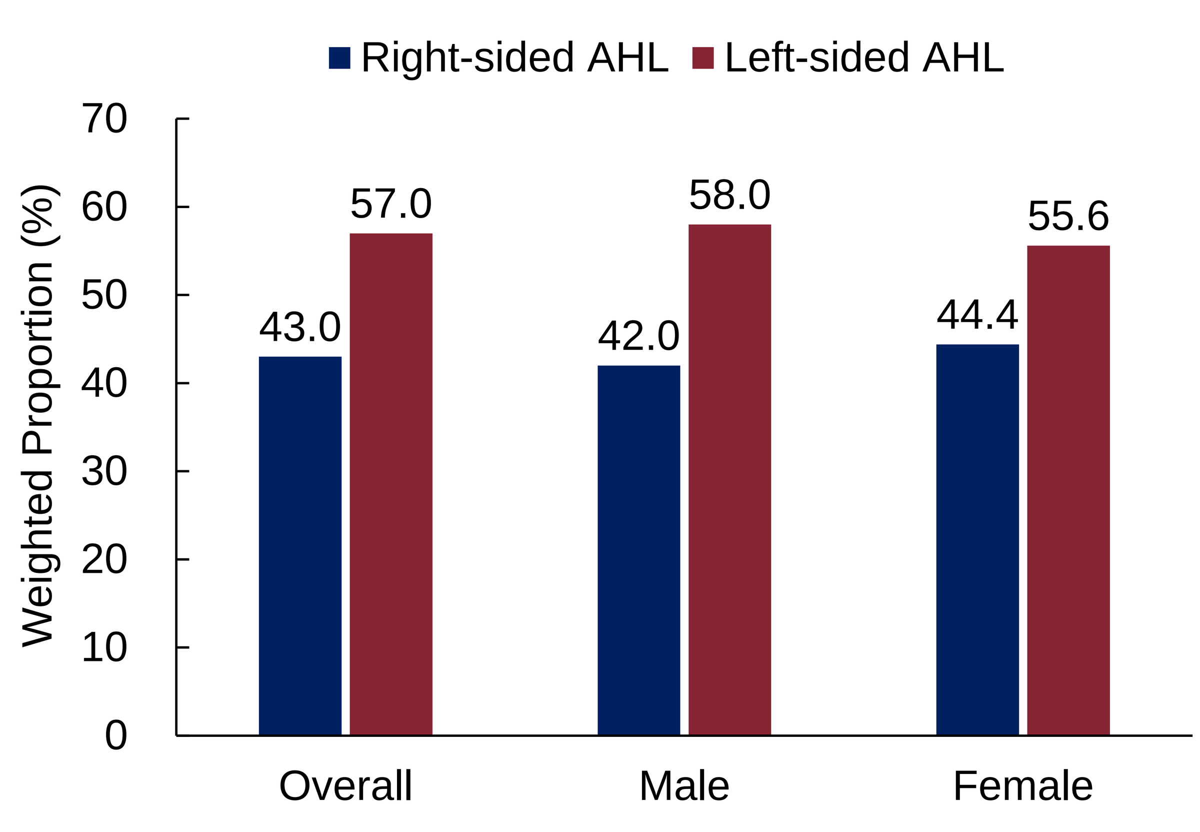

Figure 4. Left-sided AHL was more common than right-sided AHL overall and in both sexes, indicating a consistent left-sided predominance across sex subgroups.

Interaural Differences by Age (n = 447)

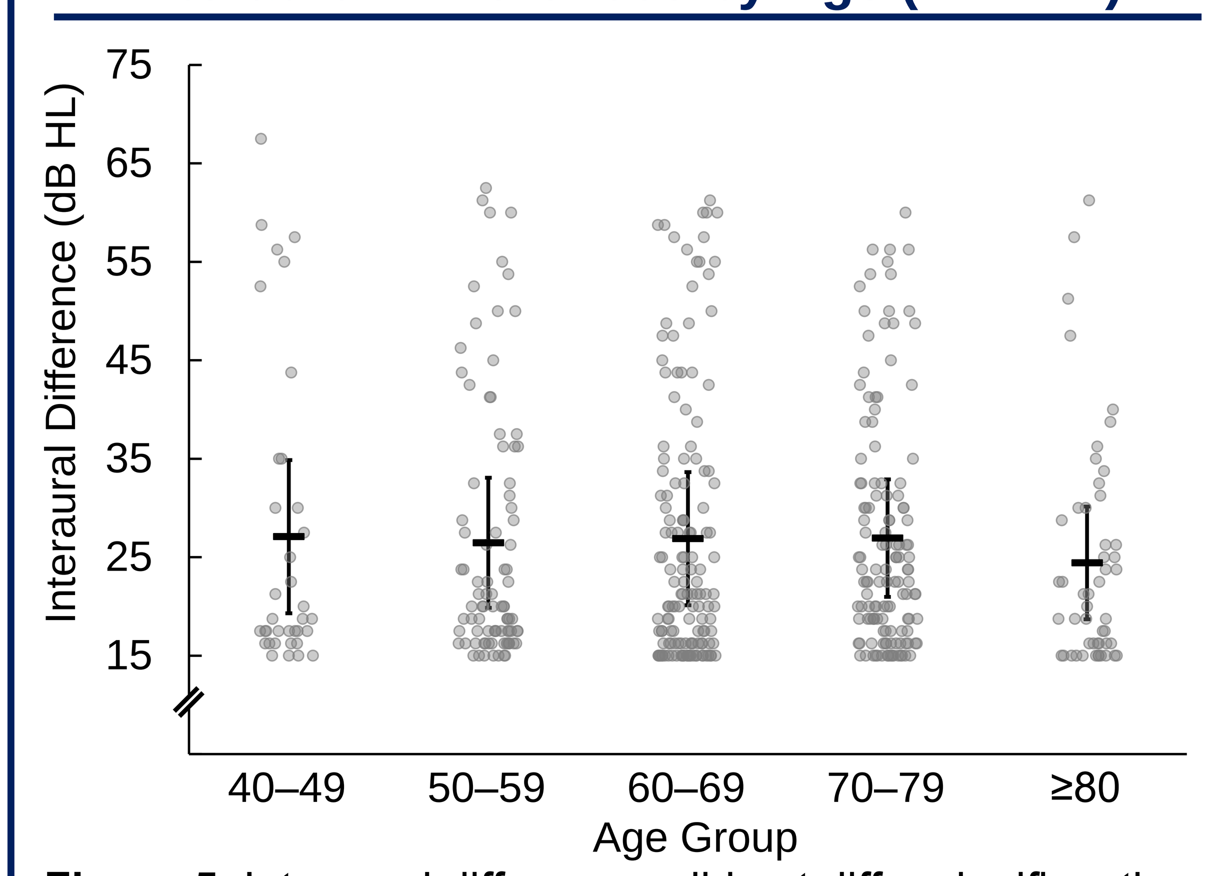

Figure 5. Interaural differences did not differ significantly across age groups (p = .530), with most cases clustering within the 15–20 dB HL range. Horizontal bars and error bars indicate mean ± ½ SD.

Conclusion

In Korean adults aged ≥40 years, the weighted prevalence of AHL was **6.5%**. AHL prevalence increased with age overall and was higher in men than in women. AHL was characterized by **higher-frequency asymmetry**, **left-sided predominance**, and clustering near the lower end of the asymmetry range, with **most cases falling within the 15–20 dB HL range.** These findings indicate that AHL is most often expressed as relatively mild interaural asymmetry rather than marked asymmetry.

Resources

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The authors declare no conflicts of interest.